Chap 13

Derivative :

$$r(t) = f(t)i + g(t)j + h(t)k$$

$$r'(t) = \frac{dr}{dt} = \lim_{\Delta t \to 0} \frac{r(t+\Delta t) - r(t)}{\Delta t} = \frac{df}{dt}i + \frac{dg}{dt}j + \frac{dh}{dt}k$$

If r is a differentiable vector function of t of constant length, then

$$\boxed{r \cdot \frac{dr}{dt} = 0}$$

Indefinite Integral

$$\boxed{\int r(t)\, dt = R(t) + c}$$

Ex: $\int ((\cos t)i + j - 2tk)dt = \left(\int \cos t\, dt\right)i + \left(\int dt\right)j - \left(\int 2t\, dt\right)k$

$= (\sin t + C_1)i + (t + C_2)j - (t^2 + C_3)k$

$= (\sin t)i + tj - t^2k + C$

$C = C_1 i + C_2 j - C_3 k$

Definite Integral

$$r(t) = f(t)i + g(t)j + h(t)k \quad [a, b]$$

$$\boxed{\int_a^b r(t)\,dt = \left(\int_a^b f(t)\,dt\right)i + \left(\int_a^b g(t)\,dt\right)j + \left(\int_a^b h(t)\,dt\right.}$$

13.2 Modeling Projectile Motion

Ideal Projectile Motion Equation:

$$\boxed{r = (V_0 \cos\alpha)t\,i + \left[(V_0 \sin\alpha)t - \frac{1}{2}gt^2\right]j}$$

$$\boxed{\text{Max height} \quad y_{max} = \frac{(V_0 \sin\alpha)^2}{2g}}$$

$$\boxed{\text{Flight time} \quad t = \frac{2V_0 \sin\alpha}{g}}$$

$$\boxed{\text{Range} \quad R = \frac{V_0^2}{g}\sin 2\alpha}$$

13.3 Arc Length & Unit Tangent Vector T

Length of a smooth curve

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

$$L = \int_a^b |v|\, dt$$

Arc Length Parameter with Base Point $P(t_0)$

$$s(t) = \int_{t_0}^{t} \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2 + [z'(\tau)]^2}\, d\tau$$

$$s(t) = \int_{t_0}^{t} |v(\tau)|\, d\tau$$

Unit Tangent Vector:

$$T = \frac{dr}{ds} = \frac{dr/dt}{ds/dt} = \frac{v}{|v|}$$

13.4 Curvature & Unit Normal Vector N

Curvature:

$$\boxed{K = \left|\frac{dT}{ds}\right| = \frac{1}{|V|}\left|\frac{dT}{dt}\right|}$$

Principal Unit Normal:

$$\boxed{N = \frac{1}{K}\cdot\frac{dT}{ds} = \frac{dT/dt}{|dT/dt|}}$$

13.5 Torsion & Unit Binomial Vector B

Torsion: $B = T \times N$

$$\boxed{\tau = -\frac{dB}{ds}\cdot N}$$

$$T = \frac{\begin{vmatrix} \dot{x} & \dot{y} & \dot{z} \\ \ddot{x} & \ddot{y} & \ddot{z} \\ \dddot{x} & \dddot{y} & \dddot{z} \end{vmatrix}}{|v \times a|^2}$$

Tangential & Normal components of Acceleration

$$\boxed{a = a_T T + a_N N}$$

$$\boxed{a_T = \frac{d^2s}{dt^2} = \frac{d}{dt}|v|} \quad , \quad \boxed{a_N = \kappa\left(\frac{ds}{dt}\right)^2 = \kappa|v|^2}$$

Formula for a_N:

$$\boxed{a_N = \sqrt{|a|^2 - a_T^2}}$$

Vector formula for Curvature:

$$\boxed{K = \frac{|v \times a|}{|v|^3}}$$